# TCS Coding Questions with Answers

July 3, 2019 by

## TCS Programming Questions with Answers

TCS Programming Test Questions are not like a general Coding Round Questions with Solutions it is all together different from C programming. We have analysed over 100+ TCS Programming Questions they use Command Line Programming in the Programming Section of the test. Below you will find TCS Programming Round Questions, TCS Coding Questions that are asked constantly in TCS Placement Papers Programming, programming Test of TCS Placement Coding Questions, TCS C Coding Questions, TCS C Programming Questions, TCS Coding Section, TCS C Programming Questions and Answers.

Check Questions Here



## Important Note

TCS has just introduced new test pattern for 2020 Batch pass-outs students. The pattern for 2019 pass-outs being the same. Below are the tables with detailed analysis on the same. Make sure you do not refer any other paper pattern as below one is the most updated paper pattern for both 2019 and 2020 Passouts.

**For 2020 Passouts :**

- No Of Questions:- 1 Question
- Time - 30 mins

**For 2019 Passouts :**

- No Of Questions:- 1 Question
- Time - 20 mins

Get all the questions asked Today in TCS here - TCS Ninja Live Questions Dashboard TCS Placement Papers Programming

**Online Classes**

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## TCS Paid Material –

In Online class you will get preparation paid materials access also, but if you just want materials then you can buy the ones below –

- TCS Aptitude Preparation Paid Materials
  - Basic
  - Pro
- TCS Coding Preparation Paid Materials
  - Basic
  - Pro
- TCS Verbal English Preparation Paid Materials
  - Basic
  - Pro
- TCS Programming MCQ Preparation Paid Materials
  - Basic
  - Pro

or

1. Click here to join Online Live Class

| 31 | 37 |
| --- | --- |
| MINUTES | SECONDS |

# Free Materials

## Checking if a given year is leap year or not

**Explanation:**

To check whether a year is leap or not

**Step 1:** We first divide the year by 4.

If it is not divisible by 4 then it is not a leap year.

If it is divisible by 4 leaving remainder 0

**Step 2:** We divide the year by 100

**If it is not divisible by 100 then it is a leap year.**

If it is divisible by 100 leaving remainder 0

**Step 3:** We divide the year by 400

If it is not divisible by 400 then it is a leap year.

If it is divisible by 400 leaving remainder 0

**Then it is a leap year**

C

```c
#include
int leap(int year)
{
    if(year%4 == 0)
    {
        if( year%100 == 0)
        {
            if ( year%400 == 0)
                printf("%d is a leap year.", year);
            else
                printf("%d is not a leap year.", year);
        }
        else
            printf("%d is a leap year.", year );
    }
    else
        printf("%d is not a leap year.", year);
        return 0;
}

int main()
```

```
{
    int yr, a;
    printf("Enter year : "); //enter the year to check
    scanf("%d",&yr);
    a = leap(yr);
return 0;
}
```

C++

Java

Python

## Prime Numbers with a Twist

Ques. Write a code to check whether no is prime or not. Condition use function check() to find whether entered no is positive or negative ,if negative then enter the no, And if yes pas no as a parameter to prime() and check whether no is prime or not?

- **Whether the number is positive or not, if it is negative then print the message "please enter the positive number"**
- **It is positive then call the function prime and check whether the take positive number is prime or not.**

C

```
#include
void prime(int n)
{
int c=0;
    for(int i=2;i<n;i++)
    {
        if(n%i==0)
        c = c+1;
    }
    if(c>=1)
        printf("%d is not a prime number",n);
    else
        printf("%d is a prime number",n);
}
void main()
{
int n;
printf("Enter no : "); //enter the number
scanf("%d",&n);
if(n<0)
    {
    printf("Please enter a positive integer");
    }
else
    prime(n);
```

```
}
```

C++

Java

Python

## Number Series with a Twist

Find the 15th term of the series?

0,0,7,6,14,12,21,18, 28

**Explanation :** In this series the odd term is increment of 7 {0, 7, 14, 21, 28, 35 – – – – – – }

And even term is a increment of 6 {0, 6, 12, 18, 24, 30 – – – – – – }

C

```c
#include<stdio.h>
int main()
{
   int i, n, a=0, b=0;
   printf("enter number : ");
   scanf("%d",&n);
   for(i=1;i<=n;i++)
   {
      if(i%2!=0)
      {
           a = a + 7;
      }
       else
      {
           b = b + 6;
      }
   }
      if(n%2!=0)
      {
           printf("%d term of series is %d\t",n,a-7);
      }
      else
      {
           printf("%d term of series is %d\t",n,b-6);
      }
 }
```

C++

Java

Python

## Number Series with a Twist

Link to this Question

Consider the following series: 1,1,2,3,4,9,8,27,16,81,32,243,64,729,128,2187…

This series is a mixture of 2 series – all the odd terms in this series form a geometric series and all the even terms form yet another geometric series. Write a program to find the Nth term in the series.

The value N in a positive integer that should be read from STDIN. The Nth term that is calculated by the program should be written to STDOUT. Other than value of n th term,no other character / string or message should be written to STDOUT. For example , if N=16, the 16th term in the series is 2187, so only value 2187 should be printed to STDOUT.

You can assume that N will not exceed 30.

C

```
#include<stdio.h>
int main()
{
   int i, n, a=1, b=1;
   printf("enter number : ");
   scanf("%d",&n);
   for(i=1;i<=n;i++)
   {
       if(i%2!=0)
       {
           a = a * 2;
       }
       else
       {
            b = b * 3;
       }
   }
    if(n%2!=0)
       {
           printf("\n%d term of series is %d\t",n,a/2);
       }
       else
       {
           printf("\n%d term of series is %d\t",n,b/3);
       }
}
```

C++

Java

Python

Python

## Number Series with a Twist

Link to this Question –

Consider the below series :

0,0,2,1,4,2,6,3,8,4,10,5,12,6,14,7,16,8

This series is a mixture of 2 series all the odd terms in this series form even numbers in ascending order and every even terms is derived from the previous term using the formula (x/2)

Write a program to find the nth term in this series.

The value n in a positive integer that should be read from STDIN the nth term that is calculated by the program should be written to STDOUT. Other than the value of the nth term no other characters /strings or message should be written to STDOUT.

For example if n=10,the 10 th term in the series is to be derived from the 9th term in the series. The 9th term is 8 so the 10th term is (8/2)=4. Only the value 4 should be printed to STDOUT.

You can assume that the n will not exceed 20,000.

C

```c
#include<stdio.h>
int main()
{
   int i, n, a=0, b=0;
   printf("enter number : ");
   scanf("%d",&n);
   for(i=1;i<=n;i++)
   {
       if(i%2!=0)
       {
           a = a + 2;
       }
       else
       {
            b = b + 1;
       }
   }
    if(n%2!=0)
       {
           printf("%d",a-2);
       }
       else
       {
           printf("%d",b-1);
       }
}
```

C++

Java

Python

## String with a Twist

Link to this Questions

1. The program will recieve 3 English words inputs from STDIN

    1. These three words will be read one at a time, in three separate line
    2. The first word should be changed like all vowels should be replaced by %
    3. The second word should be changed like all consonants should be replaced by #
    4. The third word should be changed like all char should be converted to upper case
    5. Then concatenate the three words and print them

Other than these concatenated word, no other characters/string should or message should be written to STDOUT

For example if you print how are you then output should be h%wa#eYOU.

You can assume that input of each word will not exceed more than 5 chars

C

```
#include
#include
int main()
{
  char a[10], b[10], c[10];
  int i,j;
  int x, y, z;
  gets(a);
  gets(b);
  gets(c);
  x = strlen(a);
  y = strlen(b);
  for(i=0;i<x;i++)
  {
      if(a[i]=='a'||a[i]=='e'||a[i]=='i'||a[i]=='o'||a[i]=='u')
      {
          a[i] = '%';
      }
  }
  for(j=0;j<y;j++)
  {
      if(b[j]=='b'||b[j]=='c'||b[j]=='d'||b[j]=='f'||b[j]=='g'||b[j]=='h'||b[j]=='j'|
         b[j]=='m'||b[j]=='n'||b[j]=='p'||b[j]=='q'||b[j]=='r'||b[j]=='s'||b[j]=='t'|
b[j]=='x'||b[j]=='y'||b[j]=='z')
      {
          b[j] = '#';
      }
```

```
  }
  z=0;
   while (c[z] != '\0') {
      if (c[z] >= 'a' && c[z] <= 'z')
       {
         c[z] = c[z] - 32;
      }
      z++;
   }
   printf("%s %s %s",&a,&b,&c);
}
```

**C++**

**Java**

## Addition of two numbers a Twist

1. Using a method, pass two variables and find the sum of two numbers.

Test case:

Number 1 – 20

Number 2 – 20.38

Sum = 40.38

There were a total of 4 test cases. Once you compile 3 of them will be shown to you and 1 will be a hidden one. You have to display error message if numbers are not numeric.

**C**

```
#include<stdio.h>
addition(int x, float y)
{
    float ans;
    ans = (float)x + y;
    printf("Answer : %.2f",ans);
}
int main()
{
   int a;
   float b;
   printf("enter first number : ");
   scanf("%d",&a);
   printf("enter second number : ");
   scanf("%f",&b);
   addition(a, b);
}
```

| C++ |
| --- |
| Java |

## Free Materials

We have a lots of free materials don't worry. Check them out below.

- Once finished with this Coding section, visit our Main TCS Dashboard here for other sections like Aptitude, Coding Questions, Command Line Programming etc.

TCS Coding Section used to be based on Command Line Argument based Coding Questions it is available no where else on the internet. We will try help you with learning command line argument based coding for TCS.

From this year there is no command Line programming but you have to use C, C++, Java, Python and Perl. We have the latest set of questions for the new pattern

**Question 0**

Find the nth term of the series.

1,1,2,3,4,9,8,27,16,81,32,243,....

```
#include<stdio.h>

#include<math.h>

int three(n)

{

int x,i;

for(i=0;i<100;i++)

{

x=pow(3,i);

if(i==n)

printf("%d",x);

}

}



int two(n)

{

int x,i;

for(i=0;i<100;i++)

{
```

```
x=pow(2,i);

if(i==n)

printf("%d",x);

}

}



int main()

{

int n;

scanf("%d",&n);

if(n%2==0)

three(n/2);

else

two(n/2+1);

}
```

**TCS Coding Questions**

# TCS Coding Questions and Solutions

**Languages allowed –**

1. C
2. C++
3. Command Line in C
4. Java
5. Perl
6. Python

Platform will be eclipse based compiler. You can code on onlinegdb.com. All your codes working on this website would run perfectly in TCS compiler.

Non Command Line Programs

1. TCS Coding Question – 1
2. TCS Coding Question – 2
3. TCS Coding Question – 3
4. TCS Coding Question – 4
5. TCS Coding Question – 5

Command Line Programs (Not in Syllabus anymore

- TCS Coding Questions – 1
- TCS Coding Questions – 2
- Area of a Triangle
- TCS Command Line Arguments – Fibonacci Series
- TCS Command Line Program to Swap two numbers
- TCS String Reversal Using Command Line Programming (Most Asked – 104 times)
- Greatest of Two Numbers using CLP

**Paid Materials**

## TCS Coding Round Questions in Test Pattern and Syllabus

- **Number of Questions** – 1
- **Total time to Solve** – 20 mins
- **Difficulty Level** – 1 Easy Questions
- **Cut-off** – Solve 1 question completely or partial output

Though command Line Programming is not there you should study it for C MCQ section as 1-2 MCQ would be there for command Line program.

## TCS C Programming Questions Basics

- Keywords like getc, scanf, getch, getchar etc can not be used.
- Instead we use command line arguments to fetch values.

## TCS Coding Section

Before reading further for TCS Coding Round Questions, I will suggest not to freak out if you don't understand in first go. Read everything once and then when you read again things will start to make sense. It is the best resource on the internet to know about TCS command line Argument Type Questions and TCS C Programming Questions and Answers.

## TCS Programming Test Facts and Questions

TCS coding test based facts and tricks, TCS Programming Questions, TCS Programming Test –

| TCS Coding Round Questions Topics | Difficulty in TCS Coding Round | Probability of asking in TCS Coding Test | Time to solve |
|---|---|---|---|
| LCM HCF TCS Coding Questions | Medium | 10 | 20 mins |
| Swapping and Reversal TCS Coding Round Questions | Hard | 20 | 20 mins |
| Factorial and Series | Medium | 20 | 20 mins |
| Operations on Numbers in TCS Coding Test | Easy | 30 | 20 mins |
| Misc | Medium – Hard | 20 | 20 mins |

## TCS Coding Questions with Answers

Let us consider this, if you wanted to write a basic C program then you would've written a main function that would've looked

like in compiler for TCS Coding Round Questions –

int Main(){

// some code

}}

## tcs placement coding questions

**However in command line arguments we write like this –**

int main(int argc, char *argv[]){

- **argc** – It is known as Argument Count and as clear from the name it stores the Count of number of Arguments.
- **argv[] –** Pointer, contains location of all the values(arguments).
- ***argv[] –** Array of values of all the arguments.
- They are parameters/arguments supplied to the program when it is invoked.

**Thus, now we have two things**

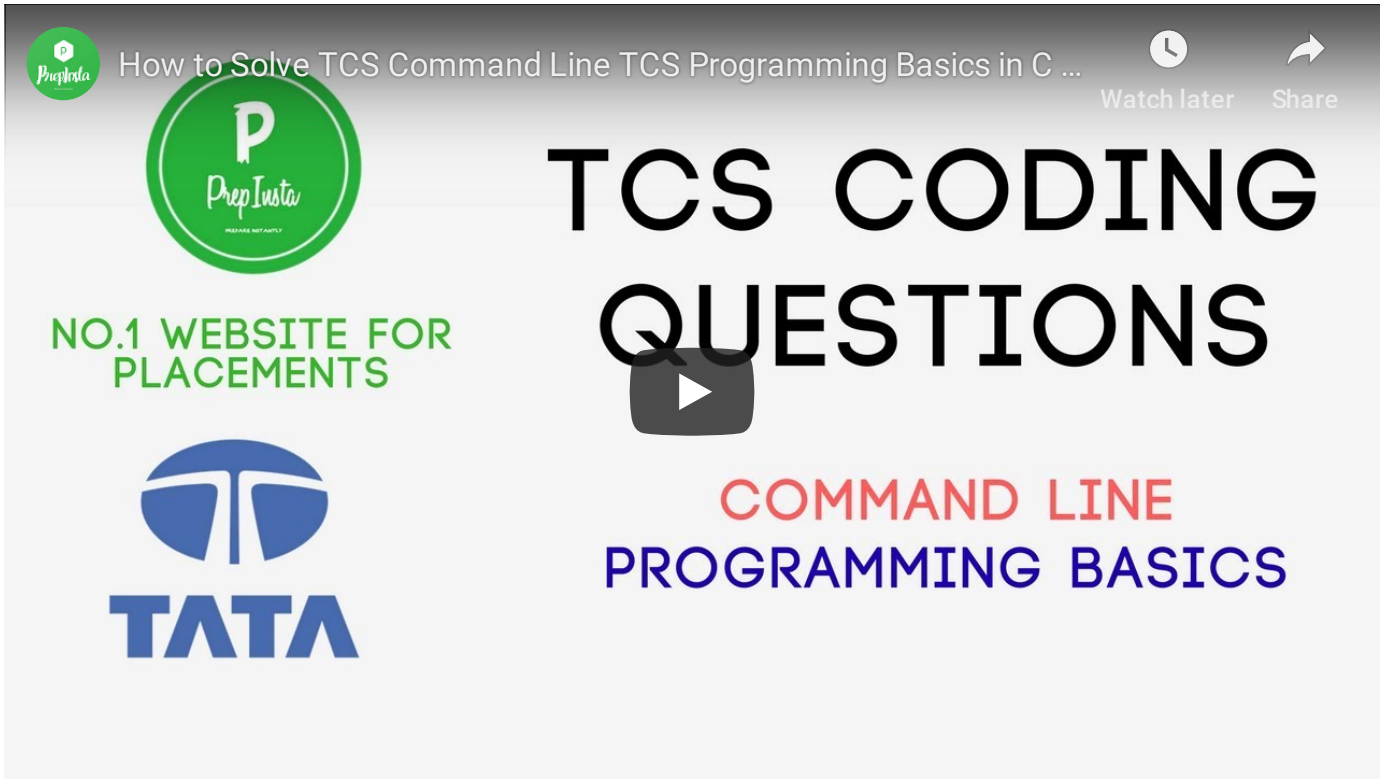
1. Total Count of number of Arguments.
2. All the values/pointer location of arguments stored in an array.

Now, you will need one more thing i.e. atoi();

- **atoi(); –** Converts string into int and atoi(argv[i]); will give the value of argument at ith location in int type format.

Now you can use an int val = atoi(argv[i]); to store the value and then print it with printf(); function.

Check out this video Below to learn how to Solve Command Line Programming.




**Quick Facts**

argv[0] contains the name, not the value so –

- All for loops must start from i=1.
- You must use the following condition

You must use the following condition:

if(argc == 1){

// do nothing since, there are no arguments, maybe ask for arguments?

}else{

// code to apply logic and print values in TCS Coding Round Questions.

}

- provided+1 +1 for file.exe
- argv[argc] is a NULL pointer.
- argv[0] holds the name of the program.
- argv[1] points to the first command line argument and argv[n] points last argument.

## tcs programming questions and Answers

| TCS Programming Questions with Answers Type | No of times asked in TCS Programming Test | Importance for Programming Round | Difficulty in TCS C Programming Questions |
|---|---|---|---|
| TCS C Programming Questions on String Reversal | 14 | Low | Very High |
| Average of two Numbers | 39 | Medium | Medium |
| TCS Placement Papers Programming LCM of two Numbers | 51 | High | Medium |
| HCF/GCD Question | 66 | High | Medium |

Command Line Arguments here – Read this properly.

```
// Program to print all value of
// command line argument
// once we get the value from command
// line we can use them to solve our problem.
#include  // this is used to print the result using printf
#include  // this is used for function atoi() for converting string into int
```

// argc tells the number of arguments
// char *argv[] is used to store the command line
arguments in the pointer to char array i.e string format
int main(int argc, char *argv[])
{
// means only one argument exist that is file.exe
if (argc == 1) {
printf("No command line argument exist Please provide them first \n");
return 0;
} else {
int i;
// actual arguments starts from index 1 to (argc-1)
for (i = 1; i < argc; i++) {
int value = atoi(argv[i]);
// print value using stdio.h library's printf() function
printf("%d\n", value);
}

```
return 0;
}
}
```

## TCS Programming Test Based FAQ's

**Ques. What is the level of difficulty of the question asked in TCS programming round?**

Ans. They are easy but you need to have good grip over basic C/C++ input/output or pattern based programs.

Logics like Recursion, For loops, If loops etc

**Ques. What languages can we use in TCS Coding Round Questions?**

Ans. In TCS Coding Round Questions currently you can only use C/C++ and Java for coding in TCS based Compiler that is mostly GCC type of compiler.

**Ques. Is PrepInsta enough to prepare for TCS Coding Round and Questions asked in the exams?**

Ans. Yes, it is the best resource out there in the internet to prepare for TCS Coding section paper.

## Rules for TCS Coding Round Questions Section:

There is only one question for 20 minutes.
• It has 10 attempts(We can compile only 10 times).
• We must start our code from the scratch.
• The coding platform is divided into two, one for writing the code and other for output. We should write the whole program.
• We can't use any input functions like scanf(), getch(), getchar().
• The input to be provided should be read as command line arguments.

We must only print exact output.
• Output must not be re-framed by extra words.
• If there is any error, the error will be shown in the output dialog box.
• The errors are clearly mentioned.
• If there are no errors, a message like "compiled successfully" will be printed.
• Along with that they will mention four test cases are 'passed' or 'failed'. They are indicated like private and public test cases. They have not mentioned what is the test case, which is difficult to understand.

Don't Compile again and again since compiler takes 25 seconds and each time you compile 25 seconds will become lesser in the time you have to code in TCS Placement Papers Programming.

**Ques. How to clear the TCS coding round?**

Ans. First you must learn command line programming from our coding dashboard and then try sample questions given on the dashboard to know how to clear the tcs coding round  and TCS C Programming Questions and Answers

**Ques. What are some other websites where we can find more questions?**

Ans. You can find more questions on MyGeekMonkey website here – http://mygeekmonkey.com/tcs-coding-questions.html

| Summary | |
|---|---|
| **Reviewer** | Shourya |
| **Review Date** | 2017-11-21 |
| **Reviewed Item** | TCS Coding Questions |
| **Author Rating** | ★★★★★ |

# 21 comments on “TCS Coding Questions with Answers”

-  *PrepInsta* says:
  November 21, 2017 at 10:24 pm
  Tcs Coding question

  Reply
-  *Venny* says:
  August 20, 2018 at 2:10 pm
  This year there is command line argument programs or normal c programs

  Reply
  -  *Nikhita* says:
    August 28, 2018 at 10:25 pm
    Only normal standard input or output type questions

    Reply
-  *Nabanita Bose* says:
  August 20, 2018 at 9:41 pm
  SORRY SIR..But This is very illogical that we have to pay first ,then we can have the coding questions and answers.We are following your site because its too helpful and easy to understand.Here we came to learn. I don’t want the answers but atleast questions should be available for students,So we can try out by ourselves.It isn’t possible for every students paying first and get the papers.If it is paid Then there is no need of following your site.if my words been rude , i am extremely sorry. □

  Reply
-  *Abhi Parker* says:
  May 27, 2019 at 11:39 pm
  Q. 1
  public class Main
  {
  public static void main(String[] args) {
  int no=9;
  if(no%2==0){
  int res=6*(no-1)/2;
  System.out.println(res);
  }else{
  int res=7*(no/2);
  System.out.println(res);
  }
  }
  }

  Reply
-  *Abhi Parker* says:
  May 27, 2019 at 11:48 pm

Q. 2
public class Main
{
public static void main(String[] args) {
int no=10;

if(no%2==0){
double res=Math.pow(3,((no/2)-1));
System.out.println((int)res);
}else{
double res=Math.pow(2,((no/2)));
System.out.println((int)res);
}
}
}

Reply

-  *Tanaya* says:
June 26, 2019 at 10:45 pm
solution-question 0
#include
int main()
{
int no;
scanf(“%d”,&no);
if(no%2==1)
{
int term=(no+1)/2;
int res=7*(term-1);
printf(“%d”,res);
}
else
{
int term=(no/2);
int res=6*(term-1);
printf(“%d”,res);
}
return 0;
}

Reply

-  *mụn* says:
June 27, 2019 at 12:27 am
sẹo rỗ là một loại sẹo nhiều ở nhiều
người, tuy nhiền không phải ai cũng biết cách điều trị sẹo rỗ hiệu quả nhất, an toàn nhất.

Ngày nay sẹo rỗ tuy không gây ra nguy hiểm hay ảnh hưởng đến tình trạng sức khỏe nhưng nó làm mất đi tính thẩm mỹ trên làn da,
khiến nhiều người e ngại, không tự tin khi
giao tiếp.

Reply

-  *KOosY* says:
  June 27, 2019 at 6:21 pm
  Q2

  #include
  #include
  int cubic(int a);
  int sq(int b);
  int n,i=0,j=0,f,o;
  void main(){

  scanf(“%d”,&n);
  if(n%2==0){
  int d = cubic(n);
  printf(“%d”,d);
  }
  else{
  int e = sq( n);
  printf(“%d”,e);

  }

  }
  int cubic(int a){
  int g = (a/2)-1;
  for(i=0;i<=g;i++){
  f=pow(3,i);
  }
  return f;

  }

  int sq(int b){
  int h = (b-1)/2;
  for(j;j<=h;j++){
  o = pow(2,j);

  }
  return o;

  }

  Reply
-  *KOosY* says:
  June 27, 2019 at 6:30 pm
  Q4

  #include
  #include

  void main(){
  int n;
  scanf(“%d”,&n);
  if(n%2==0){
  printf(“%d”,(((n-2)/2)));
  }

else{
printf(“%d”,(n-1));
}

}

Reply

-  *KOosY* says:
June 28, 2019 at 12:13 pm
Q0

#include
#include

void main(){
int n;
scanf(“%d”,&n);
if(n%2 == 0){

printf(“%d”,(((n/2)-1))*6);

}
else
printf(“%d”,( ((n-1)/2)*7 ));

}

Reply

-  *https://mcknighthoward50.tumblr.com/post/185684346321/how-to-get-your-penis-bigger-without-male* says:
June 29, 2019 at 11:02 pm
Whats up this is kind of of off topic but I was wanting to know if blogs use
WYSIWYG editors or if you have to manually code with HTML.
I’m starting a blog soon but have no coding knowledge so
I wanted to get guidance from someone with experience. Any help would be enormously appreciated!

Reply

-  *Ankit* says:
June 30, 2019 at 2:38 am
Is it necessary to clear coding round to get selected in tcs?

Reply

-  *http://www.campuslivingvillages.co.uk/tinyurl/bionativeketoprice576292* says:
June 30, 2019 at 9:26 am
I think other web-site proprietors should take this website as an model, very clean and magnificent user genial style and design, as well as the content.
You are an expert in this topic!

Reply

-  *AbhiSha* says:
June 30, 2019 at 6:08 pm